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Ensemble coding of average length and average orientation are correlated

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Abstract

Viewers can summarize redundant features in groups of objects into an ensemble percept. There appears to be separate mechanisms underlying ensemble perception of low- and high-level visual features, but it is unclear whether ensemble perception of different low-level features is supported by common mechanisms. Yörük and Boduroglu, in *Attention, Perception, & Psychophysics* 82 (2020) 852-864, investigated whether length and orientation summarization tap common mechanisms by examining the correlation between errors on length- and orientation-averaging tasks and concluded that because they did not find any correlations, the two features are summarized by different, feature-specific mechanisms. However, their study was conducted with a small sample size and included sources of individual performance variance that may have diminished correlations. We report two studies that tested the correlation between performance in the length- and orientation-averaging tasks, with larger samples and modifications that sought to reduce the sources of variance. Study 1 used ensembles that varied in both feature dimensions and Study 2 used ensembles that only varied in the task-relevant dimension. Both studies showed that errors in length- and orientation-averaging are correlated, suggesting that ensemble perception of these low-level features is supported, at least to some extent, by a common ability.

Keywords: ensemble perception, mean length, mean orientation, individual differences

1. Introduction

The visual world is structured and contains groups of similar objects, like individual faces in a crowd or blades of grass in a field. Faced with limitations due to eye movements (Inamdar & Pomplun, 2003), working memory capacity (Agam & Sekuler, 2007; Cohen et al., 2016; Luck & Vogel, 2013), attentional resources (Ghorashi et al., 2007; Wolfe et al., 2007) and visual crowding (Levi, 2008; Millin et al., 2014), one way to simplify complex scenes is to summarize redundant information into an ensemble percept (Whitney & Yamanashi Leib, 2018). When the features of individual items are not important, ensemble perception can be a fast (Ariely, 2001; Haberman & Whitney, 2007) and efficient (Alvarez, 2011) way to deal with a large number of items. For example, if a speaker wants to know how receptive a crowd is to their presentation, ensemble perception of faces in the audience can produce a gist sense of the average emotional expression, without the need to attend to each face individually.

Viewers can extract summary information from both low- and high-level features, and some have proposed that different mechanisms are devoted to processing each type. Viewers can extract low-level statistical summary information from ensembles about simple features such as item size (Ariely, 2001; Im & Halberda, 2013; Yörük & Boduroglu, 2020), orientation (Haberman et al., 2015; Yörük & Boduroglu 2020), or color (Haberman et al., 2015; Maule et al., 2014). As evidence of more complex feature dimensions, viewers can make ensemble judgments about face identity (Neumann et al., 2018; Yamanashi Leib et al., 2014) and emotion (Elias et al., 2017; Haberman et al., 2009). Importantly, errors in ensemble perception are correlated within low-level (e.g., orientation and color) and within high-level feature dimensions (e.g., facial identity and expression), suggesting abilities that generalize across at least some differences in stimuli (Haberman et al., 2015). Meanwhile, there appears to be at least two different abilities, as errors in ensemble perception of low-level features are not correlated with errors in ensemble perception of high-level features (Haberman et al., 2015).

Putting aside the distinction between abilities for low- and for high-level features, the generality of a low-level ensemble coding mechanism has been questioned. Yörük and Boduroglu (2020, hereafter YB2020) compared errors in judgments of average length and average orientation and found Bayesian evidence supporting no correlation, arguing against a common mechanism for ensemble processing of these low-level features. YB2020 concluded there must be multi-level, feature-specific mechanisms for ensemble perception. Here, we revisit whether performance in judgments for average length and orientation are related because of concerns regarding limitations of the YB2020 study.

The main concern is that of sample size. YB2020 included three studies with 21, 24 and 37 participants. Using null hypothesis statistical testing (NHST), with $N=26$, a correlation would need to reach .52 to be detected with 80% power. With $N=40$, that correlation would have to be .43. The largest correlation observed between errors for length and orientation judgments in that study reached .28. YB2020 used Bayesian statistics to evaluate support for their effects. A common threshold used to label evidence as more than inconclusive in favor of the null hypothesis is a $BF_{01} > 3$ (Jeffreys, 1961). The strongest evidence for a null correlation in their study was a BF_{01} of 3.52, and the BF s in the experiment with 40 participants (37 participants analyzed) were both inconclusive (i.e., $BF_{01} < 3$). Accordingly, a study with larger sample sizes may afford a more conclusive test, and better precision to estimate the effect size.

We also aimed to improve on the stimuli and design used in YB2020. YB2020 included some lines so short (as short as 24 pixels, compared to our shortest length of 44 pixels) as to make orientation perception difficult, thereby making ensemble judgments of orientation less consistent when viewing ensembles containing these short lines. In any display, participants could be more likely to miss short outlier lines than long outlier lines, making ensemble judgments of length less consistent and thereby deflating the correlation between performance on the two tasks. Another concern was the use of a large range of orientations (60° to 90°). With such a range, the circular mean (Mardia, 1972) can be quite different from the linear mean¹, and this ambiguity could introduce error in participants' average orientation judgments that could reduce correlations. We also avoided using mixed-task blocks (YB2020 used mixed- and single-task blocks with similar results) to avoid any potential influence of task switching. Switching between tasks can reduce performance due to increased demands on attention and cognitive control (Longman et al., 2014; Yeung et al., 2006). Because individual differences in attentional and cognitive control could have introduced variance in the mixed-task blocks and reduced correlations, we only included single-task blocks in our study. Finally, we used a fixed order of trials instead of an order randomized for each participant, to avoid confounding order effects with the measurement of abilities². Using a fixed, random order of trials is common practice in individual difference studies (e.g., Aslan & Bäuml, 2011; Ecker et al., 2010; Haberman et al., 2015).

A growing number of studies investigates the structure of abilities for ensemble perception. The default assumption, based on similar efforts in cognitive skills (Spearman, 1961) and object recognition (Richler, 2019) is of an underlying ability that contributes across all ensemble tasks. It is difficult to test all possible correlations in the large space of tasks and domains. But it is easier and especially informative to scrutinize any claim of strong independence in this space, as such claims point to possible principles governing underlying abilities. For instance, YB2020 interpreted the lack of interference in two different ensemble judgments (length/orientation averaging) in the mixed-task block as evidence supporting independent mechanisms. However, the lack of interference could be interpreted otherwise without the support of the absence of correlation. For example, participants might have represented an average bar with both features (length, orientation) regardless of whether they knew which feature to report (single-task block) or not (mixed-task block). Presence/absence of the correlation between the two ensemble abilities could support or contradict this hypothesis. Robust measurement of correlations among ensemble abilities is important for interpreting the

¹ In a circular space, such as an orientation space, mean is defined as the sum of orientation vectors of unit length divided by the number of vectors. The circular mean is different from the linear mean, and this difference between circular and linear means could be substantial when the individual values span a broad range of around half of the circular space. For instance, in an orientation space spanning the range of 0° - 180° , the linear mean of 0° and 90° is 45° but the circular mean of 0° and 90° is unspecified (neither 45° nor 135°); the linear mean of 0° , 0° , and 90° is 30° but the circular mean of 0° , 0° , and 90° is 0° (neither 30° nor 150°).

² In an experimental study where the contrast between conditions or tasks is of interest, task order is typically randomized or counterbalanced to avoid introducing a confound. In an individual differences study, however, we do not interpret mean differences between tasks or conditions. Instead, an effective way to ensure that differences among participants are not due to different trial or task order is to keep these constant across all participants.

findings of other experiments claiming independence based on experimental manipulations such as task switching.

In Study 1, we tested the correlation between ensemble judgments on the two simple visual features tested in Experiment 1 of YB2020. We found decisive evidence in support of a positive correlation between ensemble perception of length and orientation. In Study 2, we addressed a possible spurious source of correlation by removing variation in the task-irrelevant feature (for instance, all lines had the same orientation when only length is relevant for the task) and again found decisive evidence in support of a positive correlation between the two low-level ensemble judgments. In sum, this work provides evidence in support of a common mechanism for ensemble perception of low-level visual features.

2. Study 1

2.1. Method

2.1.1. Participants

All participants were recruited via Amazon Mechanical Turk (MTurk). We used Bayesian analyses to assess the strength of support for or against the correlation of interest and as a stopping criterion for data collection. Theoretically, we predicted the correlation between length and orientation should be a positive correlation and so we estimated the support for a positive correlation relative to one of 0 (BF_{+0}). We used a Sequential Bayes Factor design (Schönbrodt et al., 2017) and preregistered (<https://osf.io/6atsr/>) our target sample size ($N=75$) and stopping criterion (when BF_{+0} were lower than 1/3 or larger than 3). We wanted to exclude participants who were not trying, not motivated, misunderstood instructions or experienced technical difficulty, without excluding those trying but performing very poorly. The first kind of participants could inflate correlations, while the second kind should be included for accurate estimation.

Therefore, we only included participants with performance on either task showing above-chance accuracy (i.e., proportion correct $> 1/5$) or with errors in both tasks showing evidence of sensitivity to trial difficulty. A trial's difficulty was the proportion of correct responses from above-chance participants. Each participant's performance across trials (1 for correct, 0 for incorrect) was then correlated with this estimate of difficulty. A positive correlation indicated that a participant performed better in easy trials and worse in difficult trials. We interpreted a positive correlation as evidence that a participant was earnestly trying to do the task. Participants not showing evidence of trying (correlation between trial scores and difficulty scores for corresponding trials $< .15$) were excluded. We collected data from 130 participants but excluded 49 of them. After exclusion, 81 participants were included in the final analyses (31 female) with a mean age of 37.74 ($SD = 12.42$). Participants were paid \$2.50 upon completion of the tasks. All procedures were approved by the Vanderbilt University Institutional Review Board in charge of overseeing protection of human subjects. Informed consent was obtained prior to the experiment.

2.1.2. Stimuli

The ensembles used in both tasks consisted of 12 lines of varying lengths and orientations. The lengths of the lines ranged from 44 to 116 pixels long, in 4 pixel increments, for a total of 19 lengths. We report stimulus lengths in pixels since degrees of

visual angle would be different for each participant depending on monitor size and screen resolution. The orientations of the lines ranged from 4.5° to 85.5° clockwise from the vertical meridian, in 4.5° increments, for a total of 19 orientations.

The mean lengths of the lines used in the length-averaging task were 56, 68, 80, 92, and 104 pixels long, with a standard deviation (SD) of either 8 or 12 pixels. The orientations of the lines had a mean of 31.5°, 45°, or 58.5° with a SD of either 9° or 13.5°. The full combination of possible mean lengths, length SDs, mean orientations, and orientation SDs were used to produce 120 different ensembles (5 mean lengths $\times$ 2 length SDs $\times$ 3 mean orientations $\times$ 2 orientation SDs $\times$ 2 repetitions). For each trial, the length and orientation of each line was randomly sampled from a uniform distribution and normalized so that the mean and SD of all lines match the desired mean and SD. Trials were randomized to produce one order, which was then used for all participants. Thus, all participants viewed the fixed, random set of 120 ensembles in the fixed, random order. After viewing each ensemble, participants selected the mean length from a display of five lines all with an orientation of 45° and with lengths of 56, 68, 80, 92, and 104 pixels.

The mean orientations of the lines used in the orientation-averaging task were 18°, 31.5°, 45°, 58.5°, and 72°, with a SD of either 9° or 13.5°. The lengths of the lines had a mean of 68, 80, or 92 pixels with a standard deviation of either 8 or 12 pixels. The full combination of possible mean orientations, orientation SDs, mean lengths, and length SDs were used to produce 120 different ensembles (5 mean orientations $\times$ 2 orientation SDs $\times$ 3 mean lengths $\times$ 2 length SDs $\times$ 2 repetitions). One order of trials and one set of 120 ensembles (pre-randomized) were used for all participants. After viewing each ensemble, participants selected the mean orientation from a display of five lines all with a length of 80 pixels and with orientations of 18°, 31.5°, 45°, 58.5°, and 72°.

In both tasks, the lines in each ensemble were randomly distributed across a 5-by-4 grid of possible locations, with the exact positions jittered randomly. To prevent lines from being clustered coincidentally, 3 lines were placed in each of the 4 rows.

2.1.3. Procedure

For each trial in both tasks, participants viewed an ensemble of lines and judged the average length or average orientation (Figure 1). All ensembles were displayed for 200 ms to prevent participants from processing each line individually. Ensembles were preceded by a fixation square sequence lasting 1500 ms (1000 ms open square, 500 ms black square). Participants were then prompted to estimate the average length/orientation of the ensemble by clicking on one of the 5 buttons, each depicting a line having specific length/orientation. Five different lengths (56, 68, 80, 92, and 104 pixels) and orientations (18°, 31.5°, 45°, 58.5°, and 72°) were used in all trials for all participants. We chose these values based on separate pilot experiments, to avoid ceiling or floor performance for the two tasks. There was no time limit to make average length/average orientation judgments. Before each task, participants completed 10 practice trials with feedback to familiarize them with the tasks and with how to compute the average length or orientation of an ensemble. In the practice trials, the mean length/orientation of ensembles was one of the 5 values used in the main tasks, but the SD was 4 pixels or 4.5°. This was intended to make the tasks easy to learn. Each participant completed 120 trials of the length-averaging task followed by 120 trials of the orientation-averaging task.

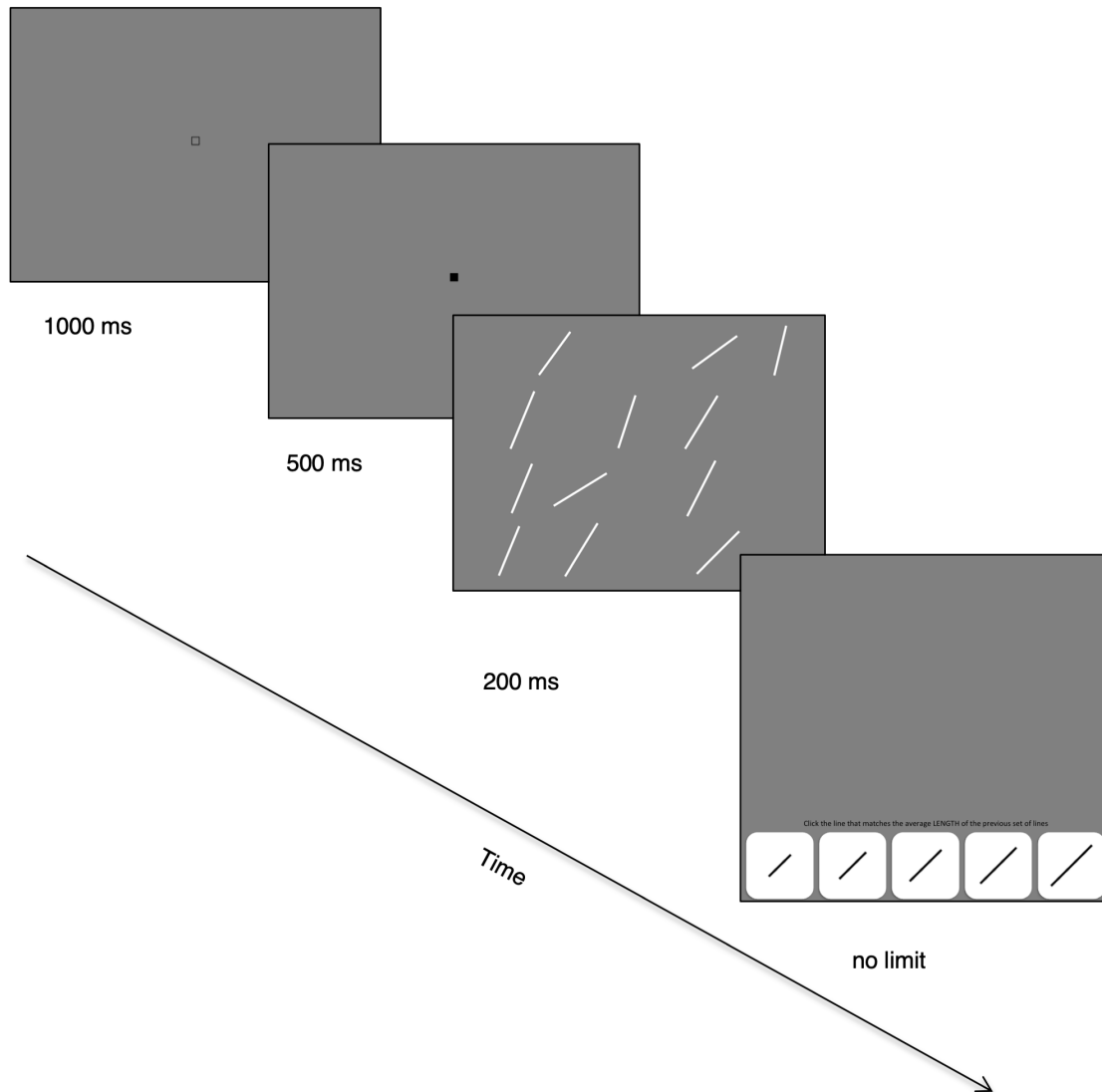
To record participant responses, we used a 5-alternative forced choice (5-AFC) task. Though this differs from the slider adjustment method used in YB2020, we selected

the 5-AFC task as a procedural compromise and to benefit from our experience running online experiments on MTurk³. This response procedure based on 5 options reduces the sensitivity of error measures relative to a continuous adjustment method. However, noisier responses should only reduce correlations between the tasks. It should be noted that the apparent precision of continuous responses may be overestimated, as responses likely include variability related to the motor and perceptual challenge of adjusting a slider to match one's mental representation. We decided to use 5 choices because we wanted a range of correct responses to lower chance level and using fewer response options allows for rapid decisions with less deliberation.

³ We used an in-house online experimental platform specialized for experiments based on n-AFC click responses. This experimental platform records all click locations during an experiment. One benefit is the ability to detect strange click patterns indicative of workers who randomly clicked through an experiment without trying.

Figure 1

An example of the sequence of one trial in the length-averaging task



Note. Participants gave their answer by clicking one of the 5 options on the response display (the words in that panel, about the response options, read “Click the line that matches the average LENGTH/ORIENTATION of the previous set of lines”).

2.1.4. Analysis

We conducted Bayesian correlation analyses using JASP (JASP Team, 2020). The Bayes factor (BF) indicates the relative support for two competing models. Here we report BF_{+0} , which reflects support in favor of a positive correlation (suggesting a common mechanism for both length and orientation judgments) relative to a correlation of 0 (YB2020’s conclusion of independent mechanisms). We used Jeffreys’ (1961) criteria for interpretation of BF values in support of a positive correlation, with values between 1 and 3 being not worth more than a mention, values between 3 and 10

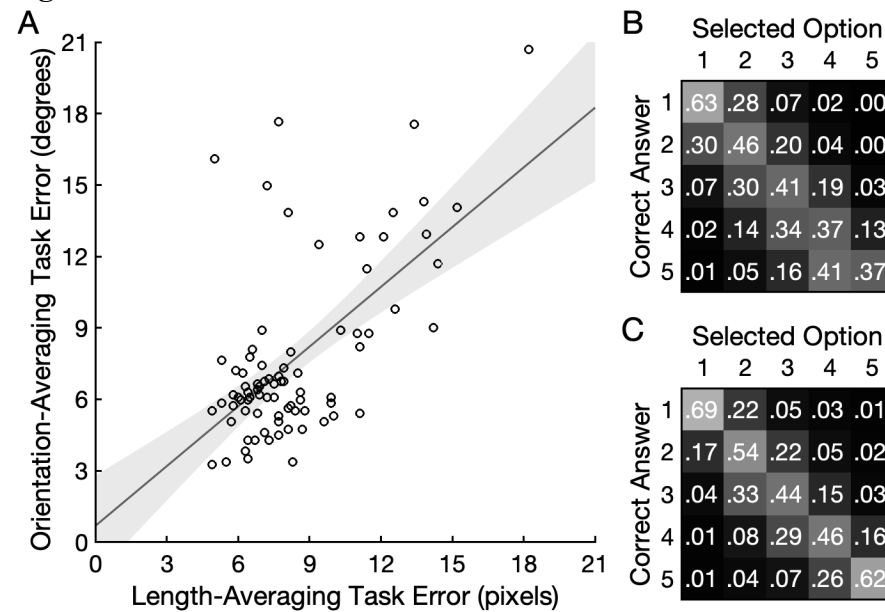
representing substantial evidence, values between 10 and 30 representing strong evidence, values between 30 and 100 representing very strong evidence, and those above 100, decisive evidence.

To be consistent with analyses done by YB2020 who used a method of adjustment, we calculated the correlation between the mean absolute error on the length-averaging and orientation-averaging tasks. We used a 5-AFC task, with each alternative corresponding to a specific length or orientation. We calculated the error, or distance, between the true ensemble mean length or orientation and the chosen length or orientation. Then, we used the average of absolute errors as our measurement and calculated the correlation between errors in the two tasks.

2.2. Results and Discussion

The average and SD of absolute error for length judgments was $M = 9.48$ pixels, $SD = 3.84$ pixels and the reliability of these measurements was $\alpha = .76$. The average and SD of absolute error for orientation judgments was $M = 9.86^\circ$, $SD = 5.27^\circ$ and the reliability of these measurements was $\alpha = .85$. We found decisive evidence in support of a positive correlation between length and orientation averaging errors, relative to the null ($r = .62$, 95% CI [.45, .73], $BF_{+0} = 2.647 \times 10^7$; Figure 2A). The correlation between the two judgments when all collected data were included was $r = .77$. This confirms our intuition that participants performing below chance on one or both tasks would only inflate the effect size – excluding them was not the cause of our rejecting the null hypothesis.

Figure 2



Note. 2A. Scatterplot showing the correlation between orientation-averaging errors and length-averaging errors. Each data point represents a participant. Linear best fit is shown with a 95% confidence interval. 2B. Confusion matrix for the length averaging task. Option 1 is the shortest line and Option 5 is the longest line. 2C. Confusion matrix for the orientation averaging task. Option 1 is the closest to the vertical line and Option 5 is closest to the horizontal line.

The reliability of each measurement places a limitation on the correlation that can be observed. The best estimate of the effect size is therefore the disattenuated correlation coefficient, whereby the observed correlation is divided by the square root of the product of the two measures' reliabilities (Muchinsky, 1996). The disattenuated correlation here is $r = .77$, suggesting that 59% of the variance in performance is shared (consistent with a common mechanism).

Figures 2B and 2C show the proportion of all participants' choices, with each row showing trials where the correct answer was 1-5 respectively. Three patterns can be appreciated from visual inspection of the confusion matrices. First, responses were roughly normally distributed in each row. Second, length-averaging task responses (Figure 2B) were biased towards choosing shorter lengths, especially when the correct answer was the longest. One explanation for this bias might be that the length-averaging task resulted in unequal distribution of attention across items (Kanaya et al., 2018), so that shorter lines were over-weighted in the averaging process. Orientation-averaging task responses do not show this pattern of bias, probably because the shortest lines could be of any orientation. Third, when the correct response was one of the extreme options (56 or 104 pixels in the length task, 18° or 72° in the orientation task, comprising 40% of the trials; hereafter extreme-mean trials), performance was high because it was not possible to make an error outside of the range of given options. To minimize the influence of the second and third patterns mentioned here, we re-analyzed our data with the extreme-mean trials removed. When extreme-mean trials were removed from analysis, we still found a positive correlation ($r = .52$, CI [.66, .33], $BF_{+0} = 5.124 \times 10^4$, disattenuated $r = .70$), which suggests that 49% of the variance in performance is shared even without the extreme-mean trials.

In addition, the pattern of results was the same when we analyzed accuracy (i.e., proportion correct, with the correct response defined as that closest to the actual average) of the 5-AFC task instead of absolute error. The average and SD of accuracy for length judgments was $M = .48$, $SD = .10$ and the reliability of these measurements was $\alpha = .76$. The average and SD of accuracy for orientation judgments was $M = .55$, $SD = .12$ and the reliability of these measurements was $\alpha = .85$. We found decisive evidence in support of a positive correlation between length and orientation averaging accuracy, relative to the null ($r = .52$, 95% CI [.33, .66], $BF_{+0} = 4.842 \times 10^4$). The disattenuated correlation here is $r = .64$, suggesting that 41% of the variance in accuracy performance is shared.

In Study 2, we seek replication and extension of our result, by repeating the same design except for the removal of task-irrelevant feature variance. That is, in Study 1, ensembles always varied in both length and orientation, even though each task required attention to only one feature. Perhaps instead of only measuring whether participants could summarize the task-relevant feature, we were also inadvertently measuring their ability to ignore the task-irrelevant feature. For instance, the orientation-averaging task performance in Study 1 could have reflected participants' abilities to ignore length, and thus correlate with participants' abilities measured in the length-averaging task. Study 2 removed this possible contribution to correlations, by varying only the task-relevant feature in each ensemble.

3. Study 2

3.1 Method

3.1.1 Participants

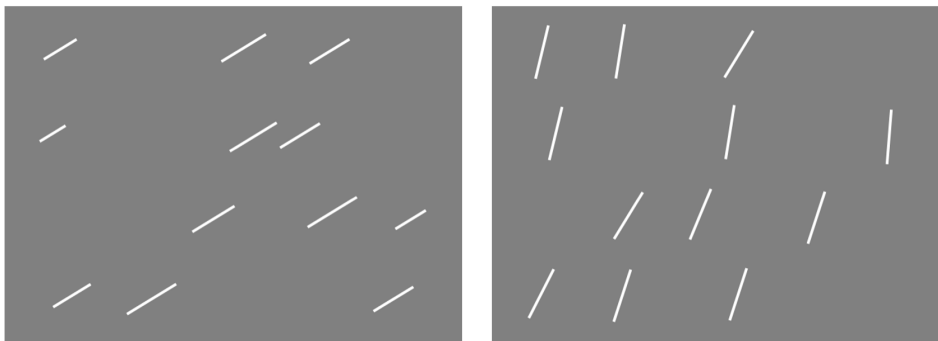
New participants were recruited via MTurk. We used the same stopping criterion as in Study 1. Our starting sample included a total of 111 participants, with 36 excluded for chance or below chance performance on one of the tasks (per our pre-registration criteria, we would not exclude a subject if performance was at chance, but we found evidence of sensitivity to trial difficulty, as determined by the correlation between trial-related errors and trial difficulties being above .15). After exclusion, 75 participants were included in the final analyses (29 female) with a mean age of 39.11 ($SD = 11.95$). Participants were paid \$2.50 upon completion of the tasks. This study was preregistered (<https://osf.io/5v49n>). All procedures were approved by the Vanderbilt University Institutional Review Board in charge of overseeing protection of human subjects. Informed consent was obtained prior to the experiment.

3.1.2. Stimuli

All parameters were the same as in Study 1, with the only exception being that the variance of the task-irrelevant dimension was always 0 (Figure 3).

Figure 3

Example ensemble displays for Study 2.



Note. The left image shows an example of a display from the length-averaging task in Study 2, where only length was varied and orientation was set at 45° for all lines. The right image shows an example of a display from the orientation-averaging task in Study 2, where only orientation was varied and length was set to 80 pixels for all lines.

3.1. Procedure

The procedure in Experiment 2 was the same as in Experiment 1.

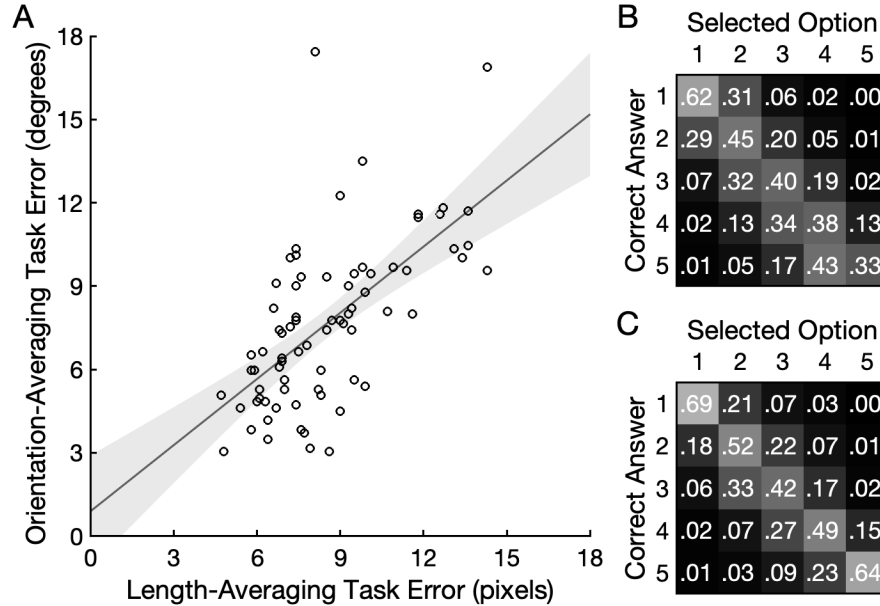
3.2. Results and Discussion

The average and SD of absolute error for length judgments was $M = 8.52$ pixels, $SD = 2.40$ pixels and the reliability of these measurements was $\alpha = .75$. The average and SD of absolute error for orientation judgments was $M = 7.70^\circ$, $SD = 2.97^\circ$ and the reliability of these measurements was $\alpha = .87$. We found decisive evidence in support of a positive correlation between length and orientation averaging errors, relative to the null ($r = .63$, 95% CI [.46, .74], $BF_{+0} = 1.893 \times 10^7$; Figure 4A). The disattenuated correlation here is $r = .79$, suggesting that 62% of the variance in performance is shared (consistent with a common mechanism).

As we found the same patterns in the confusion matrices (Figures 4B and 4C) as in Study 1, we re-analyzed the data excluding extreme-mean trials. When extreme-mean

trials (40% of the data) were removed from the analysis, we still found a positive correlation ($r = .61$, CI [.43, .73], $BF_{+0} = 3.372 \times 10^6$, disattenuated correlation $r = .84$), which suggests that 71% of the variance in performance is shared when the extreme-mean trials are removed. The extreme-mean trials in this study decreased the correlation between errors in the length and orientation averaging tasks, suggesting that a contribution from extreme-mean trials cannot explain why we observed a larger correlation than in YB2020.

Figure 4



Note. 4A. Scatterplot showing the correlation between orientation-averaging errors and length-averaging errors. Each data point represents a participant. Linear best fit is shown with a 95% confidence interval. 4B. Confusion matrix for the length averaging task. Option 1 is the shortest line and Option 5 is the longest line. 4C. Confusion matrix for the orientation averaging task. Option 1 is the closest to the vertical line and Option 5 is closest to the horizontal line.

The average and SD of accuracy for length judgments was $M = .44$, $SD = .09$ and the reliability of these measurements was $\alpha = .75$. The average and SD of accuracy for orientation judgments was $M = .55$, $SD = .12$ and the reliability of these measurements was $\alpha = .87$. We found decisive evidence in support of a positive correlation between length and orientation averaging accuracy, relative to the null ($r = .54$, 95% CI [.35, .68], $BF_{+0} = 6.300 \times 10^4$). The disattenuated correlation here is $r = .67$, suggesting that 45% of the variance in accuracy performance is shared.

4. General Discussion

In Study 1, we measured the correlation between participants' ensemble judgments on the same two simple visual features tested in Experiment 1 of YB2020. Participants viewed ensembles of lines with varying lengths and orientations and judged either the average length or average orientation. Then, we performed a Bayesian correlation analysis to examine whether errors in averaging in both tasks were correlated.

In Study 2, we repeated the procedure from Study 1 but removed the variance in the task-irrelevant feature in order to more precisely isolate length-averaging and orientation-averaging errors. In both cases, we found decisive evidence for a positive correlation between errors in the length and orientation averaging tasks, with a large disattenuated effect size, indicating that around 60% of the variance underlying ensemble processing of these two low-level features is common. While our results contradict the conclusions from YB2020, they are consistent with the results from Haberman et al. (2015) suggesting a common ability for ensemble processing of different low-level features. The current study aimed to resolve controversies from the previous studies, using a Sequential Bayes Factor design that would ensure that our results would not remain inconclusive because of insufficient data.

To do this, we focused on limitations of YB2020's study. Previous studies that argued against a common mechanism of ensemble perception of low-level features have used small samples that limit power of frequentist tests, limit the precision of effect size estimates, and can lead to inconclusive decisions with Bayesian statistics. Thus, the small sample sizes used by YB2020, combined with the sources of variance and error in their methods (using lines of short lengths, wide range of orientations, randomized trial orders for each participant) could have contributed to their small effect size. In their methods, they stated that they derived their sample size based on similar studies with similar numbers of trials that used 10 (Attarha & Moore, 2015; Poltaratski & Xu, 2013), 11 (Scolari, Vogel & Awh, 2008), 12 (Vogel & Machizawa, 2004), and 20 participants (Awh, Barton & Vogel, 2007). Power analyses are, however, specific to the contrast that is tested. Power calculations to detect a correlation cannot be based on effect sizes from group analyses and power analyses are limited when they are based on effect sizes with the very large confidence intervals associated with small samples (arguably, the lower bound of the confidence interval should be used in such a case to plan further studies). Also, power analysis to detect correlations between averaging tasks cannot be based on effect sizes for correlations reported in working memory studies, since participants' task differs (reporting one of many individual features vs. reporting the average) even though those studies used arrays of similarly many items. An alternative in Bayesian analysis is to continue collecting data until sufficient evidence is obtained (Schönbrodt et al., 2017). With modifications on the YB2020 design, and a larger sample size, we obtained conclusive support for a positive correlation between errors on the length- and orientation-averaging tasks. Although our results seem considerably different from that observed in YB2020, our correlations are not far from the upper limit of the confidence interval for their correlation. YB2020 found a correlation of $r = -.08$ with 21 participants. The 95% CI for that correlation would be $[-.49, .36]$. Our attenuated correlations of .59 and .62 are not far from the positive side of this confidence interval, and our modifications of the experimental design of YB2020 and our larger sample size may account for the difference. Our approach offers guidance for future work using individual differences in the study of ensemble judgments for different kinds of features and properties.

Given the evidence we obtained for a positive correlation between the processing of average length and average orientation, what kinds of mechanisms could subserve ensemble perception of different low-level features? One candidate is the ability to distribute attention to multiple objects effectively. Ensemble perception requires

distribution of attention across objects and can explain statistical summarization better than alternative strategies, such as focused attention to samples of a few items in an ensemble (Chong & Treisman, 2005; Chong et al., 2008). In our study, 12 lines randomly occupied 20 possible locations (in a 4-by-5 grid) and thus the locations that require cognitive resources varied from trial to trial. If a participant had a high ability to distribute cognitive resources to the lines on each trial, that participant would have performed better regardless of whether the task was about length or orientation. In turn, we could expect shared variance in performance on ensemble tasks associated with the same stimuli and with the same configuration.

We readily acknowledge two limitations of our study. First, performance in the two ensemble tasks may also have been influenced by other sources of individual differences, such as working memory capacity, sampling strategy, and ability to focus on the task-relevant feature, to the extent that these abilities contribute to our tasks. Future studies should try to estimate the proportion of variance these factors account for. This would be beneficial in the general study of any individual differences in ensemble processing. Second, our study does not directly address the results of the task-switching experiment reported by YB2020: the absence of interference associated with concurrent processing of length and orientation. In YB2020, there was no cost associated with summarizing low-level features in the mixed block trials as compared to the single block trials. The authors interpreted this finding as support for the independence of summarization mechanisms for low-level features. However, given the strong correlations we found, further studies would be needed to reconcile differences in the individual differences approach and the cognitive approach based on interference effects.

While our results suggest at least one shared mechanism involved in ensemble processing of length and orientation, we cannot be certain that all low-level features share such a mechanism. It is possible that there are some features for which ensemble processing is carried out by a feature-specific mechanism. Additionally, we can only account for up to 62% of the variance in performance for length and orientation averaging; feature-specific mechanisms for ensemble processing may account for some of the remaining variance. Our use of the 5-AFC task also could have reduced sensitivity of our measurements, leading to the underestimation of correlations, compared to using a slider to adjust a probe to the mean length or orientation, as our method limits choices to only five options. Despite this limitation but based on a large sample size that satisfied the conditions of a sequential Bayes Factor design, we found compelling evidence for shared abilities for ensemble processing of orientation and length.

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